

such as the VBS_Kakworm and VBS_Bubbleboy, viruses can now infect a user's system when the e-mail is opened.

Mixed Application/Content-based Threats

Application-based wireless threats, in which an executable program carries some malicious code, affect the receiving device. The spread of this malicious code is slow since the user must download a program with malicious code and execute the program to become infected. At the other end of the spectrum are content-based threats that spread relatively benign text messages or generate cellular phone calls. Yet, these threats can spread rapidly due to the nature of their propagation medium—entire address books of e-mails.

The third type of threat is worse than the previous two types combined. While not yet seen in the wild or even in the laboratory, a threat that integrates techniques from both of these threat types could be formidable indeed. Imagine a virus that involved the unwitting download of sophisticated malicious code attached to a shareware program that wiped out wireless device applications and propagated itself rapidly across the wireless infrastructure via address books of e-mail. Such a virus could cause damage to each device it encountered and spread across a country, or across the world, overnight. Given the reality of the ILoveYou virus and its destructive power, without adequate comprehensive wireless infrastructure virus protection, some type of highly destructive, rapidly spreading wireless virus will inevitably surface.

Threats On The Horizon To Consumers And Corporations

In many parts of the world today, cellular phones are used almost exclusively for voice communication. Yet, as cellular phone technology is merged with the platform-independent Java programming language and emerging technologies such as Bluetooth, these cell phones will be able to send and receive data and applications, even from one wireless device directly to another wireless device. The line between PDAs and cellular phones is already blurred, and few dispute that the integrated, transaction-enabled wireless device that handles both voice and data will soon become a widespread reality.